

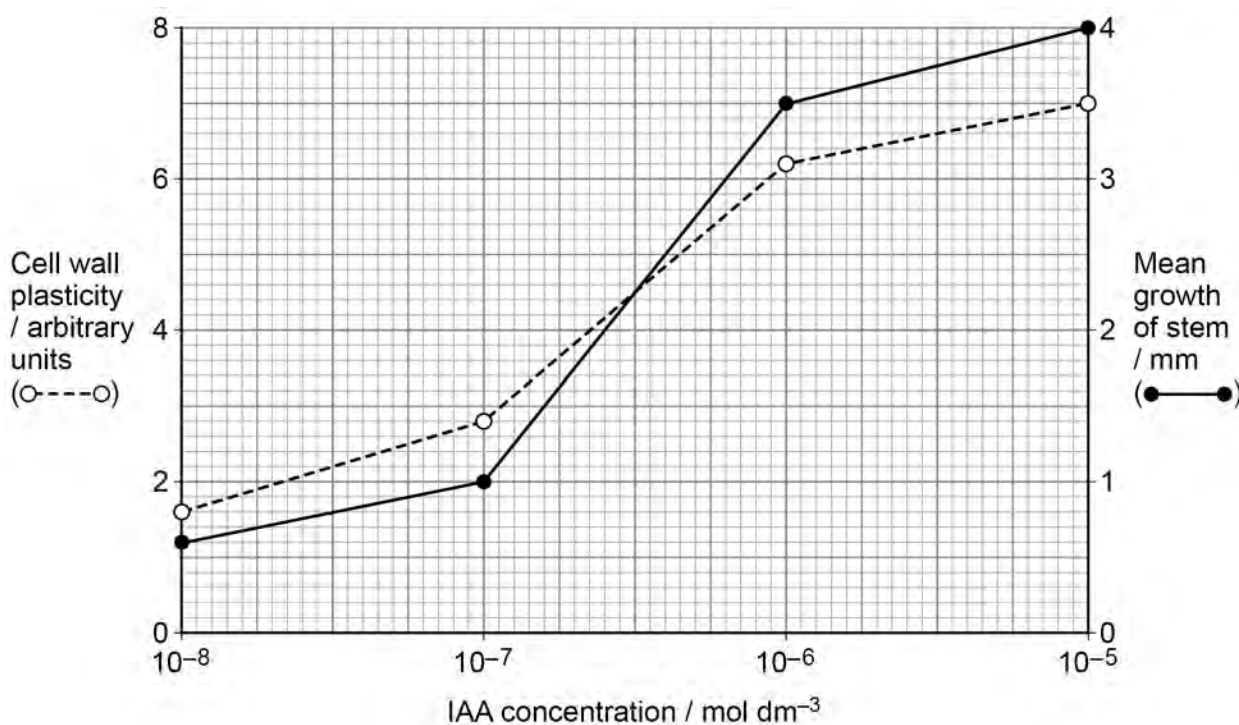
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Plant cells are surrounded by a rigid cell wall. Cell wall plasticity refers to the ability of a cell wall to stretch permanently. IAA is a growth factor which can activate enzymes that loosen the cell wall.

Scientists investigated the effect of IAA concentration on cell wall plasticity and the growth of stem segments.

Figure 6 shows some of the scientists' results.

Figure 6



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Use the information provided to explain the relationship between IAA concentration, cell wall plasticity and mean growth of the stem segments.

[2 marks]

08.2

Use **Figure 6** to calculate the percentage increase in mean growth of the stem segments when the IAA concentration was increased from $10^{-8} \text{ mol dm}^{-3}$ to $10^{-5} \text{ mol dm}^{-3}$

Give your answer to 2 significant figures.

Show your working.

[2 marks]

Answer _____ %

Question 8 continues on the next page

Gibberellic acid (GA) is another plant growth factor. A student was asked to design and carry out an investigation into the effect of different concentrations of GA on the growth of stem segments.

The student was provided with:

- 10^{-1} , 10^{-2} , 10^{-3} , 10^{-4} and 10^{-5} mol dm⁻³ concentrations of GA solution
- distilled water
- 6 Petri dishes and access to glassware
- 60 stem segments of different lengths.

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A technician produced the different concentrations of GA solution from a stock 10^{-1} mol dm⁻³ concentration of GA.

Describe how the technician produced the 10^{-3} mol dm⁻³ solution.

[1 mark]
